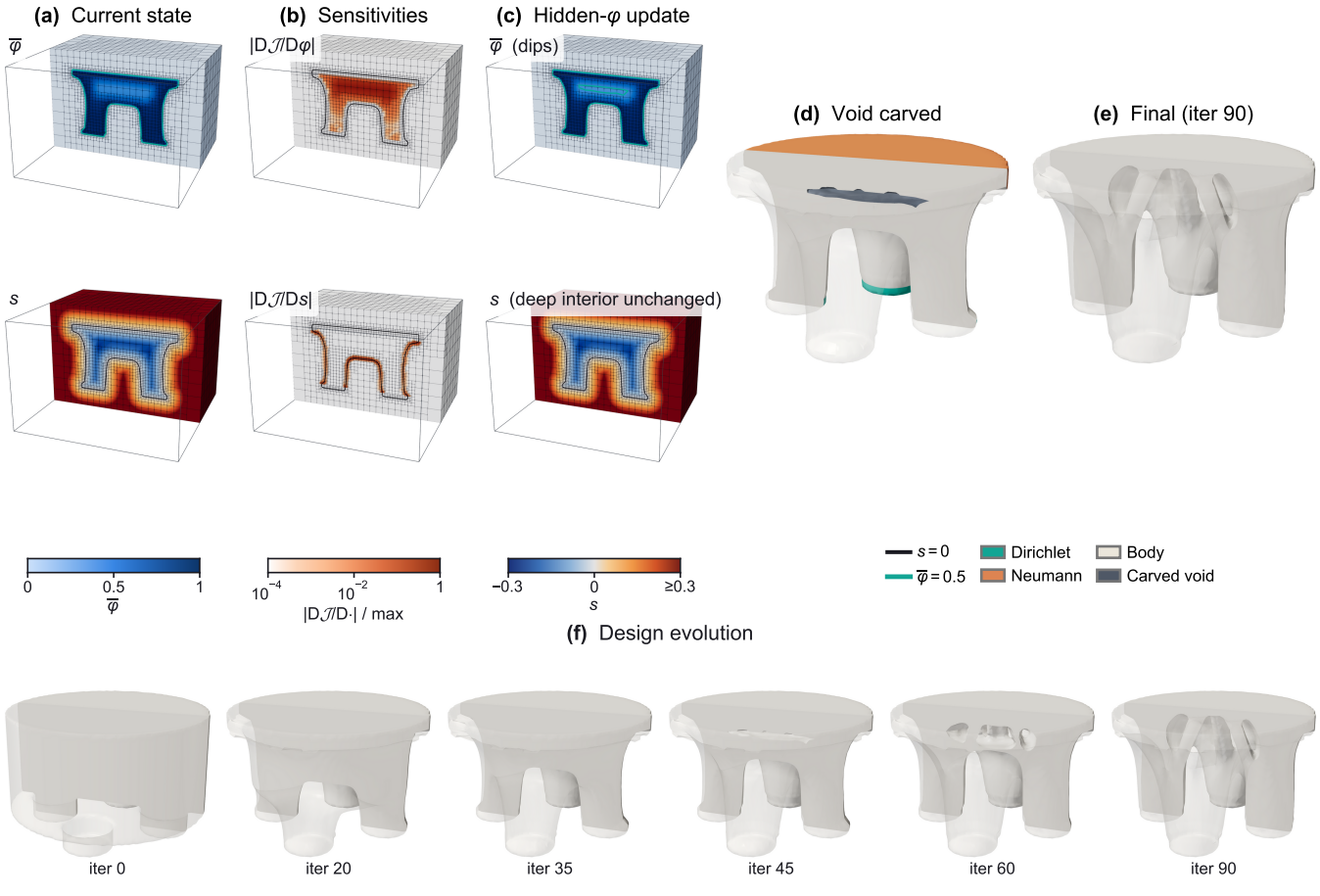

(a) Current state
(b) Sensitivities
(c) Hidden-φ update
φ̄
|D𝒥/Dφ|
φ̄ (dips)
s
|D𝒥/Ds|
s (deep interior unchanged)
(d) Void carved
(e) Final (iter 90)
0
0.5
1
φ̄
10^−4
10^−2
1
|D𝒥/D·| / max
−0.3
0
≥0.3
s
s = 0
φ̄ = 0.5
Dirichlet
Neumann
Body
Carved void
(f) Design evolution
iter 0
iter 20
iter 35
iter 45
iter 60
iter 90